

Exercise and worked solution

Financial institutions

Four parts to survey the landscape of financial institutions: a bank's margin and capital, the risk transfer of an underwriting, an insurer's actuarial mechanics, and the perverse incentive of a hedge fund's 2+20 fee. Work through each part before reading the solution.

Part A · A bank's margin and capital

PROBLEM

A bank lends on average at 5.0%, funds at 1.6% and expects to lose 0.8% of its loans. (a) Compute the NIM and what remains after expected losses. (b) Its internal model gives 6bn of capital, the standardised figure 9bn: what capital under the 2027 output floor? (c) Does equity or subordinated debt absorb losses while the bank lives?

SOLUTION

- (a) $\text{NIM} = 5.0 - 1.6 = \mathbf{3.4\%}$. After expected losses: $3.4 - 0.8 = \mathbf{2.6\%}$ for costs and profit. Expected losses are in the pricing; only unexpected losses are covered by capital.
- (b) $\text{Capital} = \max(\text{IMC}; 0.725 \times \text{SMC}) = \max(6; 0.725 \times 9) = \max(6; \mathbf{6.525}) = \mathbf{6.525bn}$. The floor at 72.5% of standardised raises the internal model.
- (c) **Equity** (going concern capital): it absorbs losses while the bank lives, solvent as long as it is positive. Subordinated debt is gone concern: it absorbs only after failure, as a cushion for depositors.

Part B · The risk transfer of an underwriting

PROBLEM

A company wants to issue 8M shares. The bank offers either a best efforts at a 1.20 USD/share fee, or a firm commitment where it buys the lot at 45 USD. Compute the bank's profit if the securities resell at (a) 50 USD, (b) 43 USD. (c) Where is the price risk in each form?

SOLUTION

- (a) At 50 USD: best efforts = $8M \times 1.20 = \mathbf{+9.6M}$; firm commitment = $8M \times (50 - 45) = \mathbf{+40M}$.
- (b) At 43 USD: best efforts = $\mathbf{+9.6M}$ (unchanged); firm commitment = $8M \times (43 - 45) = \mathbf{-16M}$.
- (c) In **best efforts**, the bank only earns a fee: price risk stays with the issuer, the bank's profit is certain but capped. In **firm commitment**, the bank bought the securities at a guaranteed price: price risk shifts to it, with a potentially far larger profit but a real chance of loss.

Part C · An insurer's actuarial mechanics and risks

PROBLEM

(a) A table gives a survival-from-birth of 0.981 at age 40 and 0.742 at age 72. Probability a 40-year-old reaches 72? (b) Which product fears longevity risk, which mortality risk? (c) A P&C insurer has a loss ratio of 68%, an expense ratio of 27%, a 1% dividend and a 2% investment income. Combined and operating ratio?

SOLUTION

- (a) $\text{Pr}(40 \rightarrow 72) = 0.742 / 0.981 = \mathbf{0.7564}$ ($\approx 75.6\%$). It is a ratio of survivals (conditional probability), not a subtraction.
- (b) **Longevity risk** (living too long) hits annuities and pensions (they pay for more years); **mortality risk** (early deaths) hits death life insurance. Being opposite, the two books partly offset.
- (c) Combined = loss + expense = $68 + 27 = \mathbf{95\%}$ (excluding investments); combined after dividends = 96% ; operating = $96 - 2 = \mathbf{94\%}$. Below 100%, the insurer is profitable even before investments.

Part D · The 2+20 fee as a call

PROBLEM

A 100M hedge fund charges 2 + 20. Compare a strategy A (a certain 3M profit) and a strategy B (50/50 between +10M and −10M). (a) The manager's expected fee in each case. (b) The investor's net return. (c) Why is the manager induced to choose B, and what safeguards exist?

SOLUTION

1. (a) Management fee = 2M throughout; incentive = $0.2 \times \max(\text{profit} - 2, 0)$. A: $0.2 \times (3 - 2) = 0.2 \rightarrow$ fee **2.2M**. B: +10 scenario $\rightarrow 0.2 \times 8 = 1.6$; −10 scenario $\rightarrow 0$; expected incentive = 0.8 $\rightarrow$ fee **2.8M**.
2. (b) Investor net: A = $3 - 2.2 =$ **+0.8M**; B = 0 (expected gross) $- 2.8 =$ **−2.8M**.
3. (c) The incentive is a **call**: the manager captures 20% of the upside without bearing the downside, and a call's value rises with volatility. B thus pays the manager more ($2.8 > 2.2$) while destroying value for the investor (gross expectation $0 < 3$). The manager is induced to take risk against the client's interest. Safeguards: **hurdle rate, high-water mark, clawback**.

Key takeaways: NIM = loans – funding cost (expected losses in pricing, unexpected in capital); output floor = 72.5% of standardised; best efforts (price risk with issuer) vs firm commitment (with bank); conditional survival = ratio; mortality (life) vs longevity (annuities/DB); combined ratio excludes investments, operating includes them; and the 2+20 fee is a call that induces risk, tempered by hurdle/high-water/clawback.